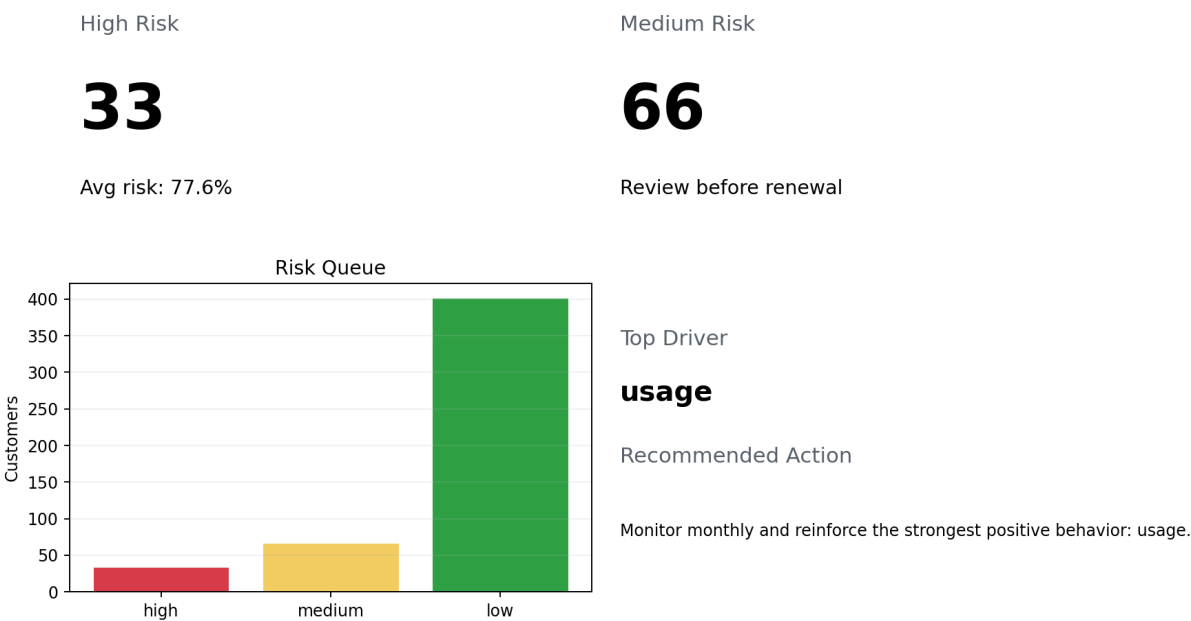


Churn Prediction + Retention Report

Executive summary: this report ranks customers by churn risk and turns model output into a prioritized retention action queue.

The model ranks customers by churn risk. The top-risk segment should be reviewed first because it concentrates the highest expected churn probability.

Churn Retention Executive Snapshot



Dataset Summary

Metric	Value
rows	500
columns	10
observed_churn_rate	23.00%
duplicate_customer_ids	0
leakage_warnings	0

Model Metrics

Metric	Value
roc_auc	0.77
pr_auc	0.5164
baseline_pr_auc	0.2267
accuracy	0.7667

Metric	Value
precision	0.4615
recall	0.1765
f1	0.2553
confusion_matrix	[[109, 7], [28, 6]]

Risk Segment Breakdown

risk_segment	customers	avg_churn_probability
high	33	0.7761
medium	66	0.5384
low	401	0.1243

Lift Table Interpretation

The highest-risk 10% captures 39.1% of observed churners. The highest-risk 20% captures 64.3%. Review the 33 high-risk customers first.

decile	customers	churners	churn_rate	avg_churn_probability	cumulative_churners	cumulative_churn_capture_rate
1	50	45	0.9000	0.7376	45	0.3913
2	50	29	0.5800	0.4930	74	0.6435
3	50	19	0.3800	0.3146	93	0.8087
4	50	9	0.1800	0.2080	102	0.8870
5	50	5	0.1000	0.1554	107	0.9304

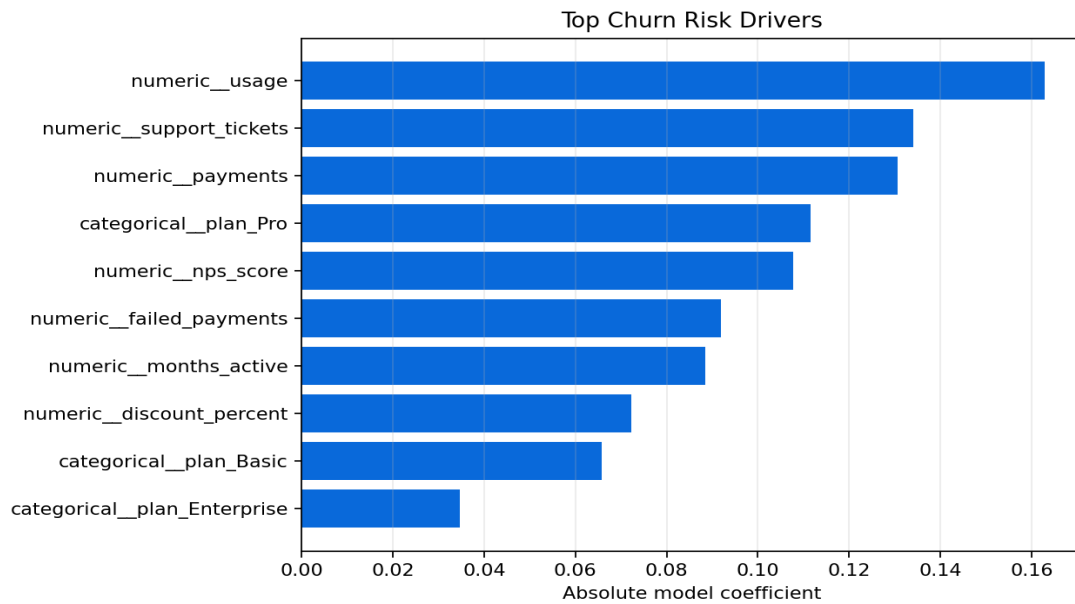
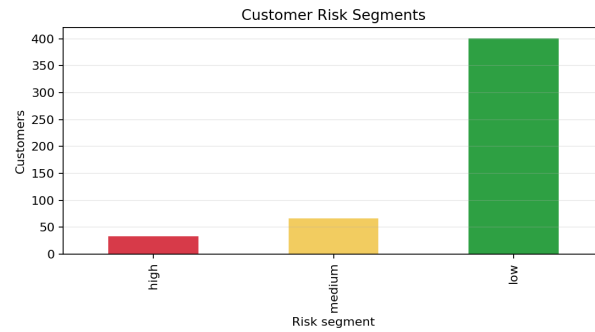
Top Recommended Actions

segment	churn_probability	likely_drivers	recommended_action
um	0.0199	usage, support_tickets	Monitor monthly and reinforce the strongest positive behavior: usage.
	0.4832	usage, payments, plan	Send targeted intervention before renewal risk increases. Suggested playbook: Offer onboarding, usage review, or workflow training to improve
	0.0396	usage, support_tickets	Monitor monthly and reinforce the strongest positive behavior: usage.
um	0.0696	usage, support_tickets	Monitor monthly and reinforce the strongest positive behavior: usage.
	0.4124	usage, payments, plan	Send targeted intervention before renewal risk increases. Suggested playbook: Offer onboarding, usage review, or workflow training to improve

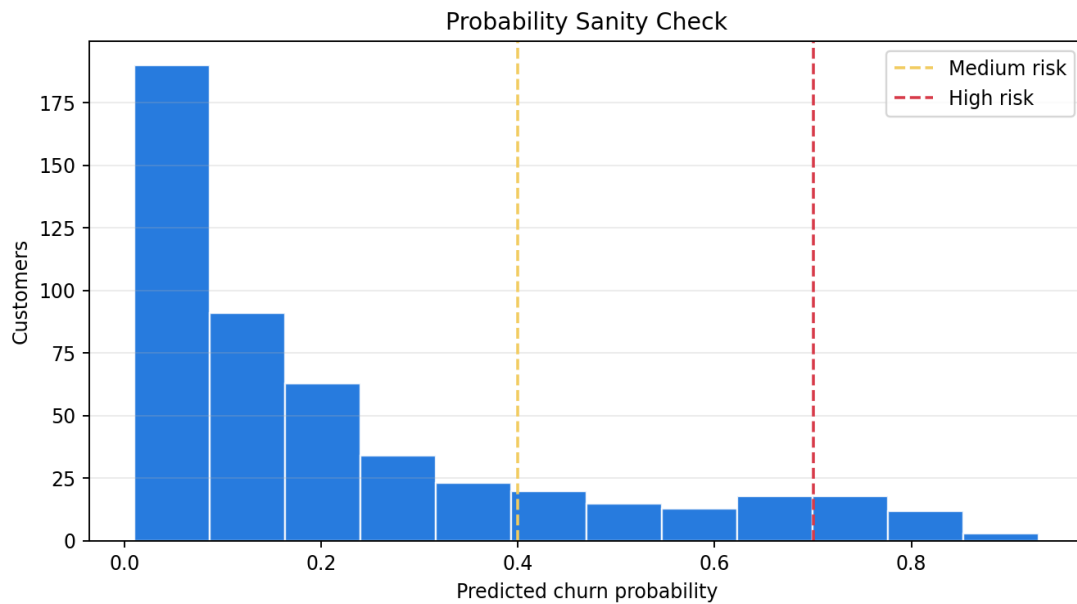
Recommended Retention Strategy

Use the retention recommendation CSV as the operating queue. High-risk customers should receive direct outreach; medium-risk customers should receive targeted education, usage review, or plan-fit checks.

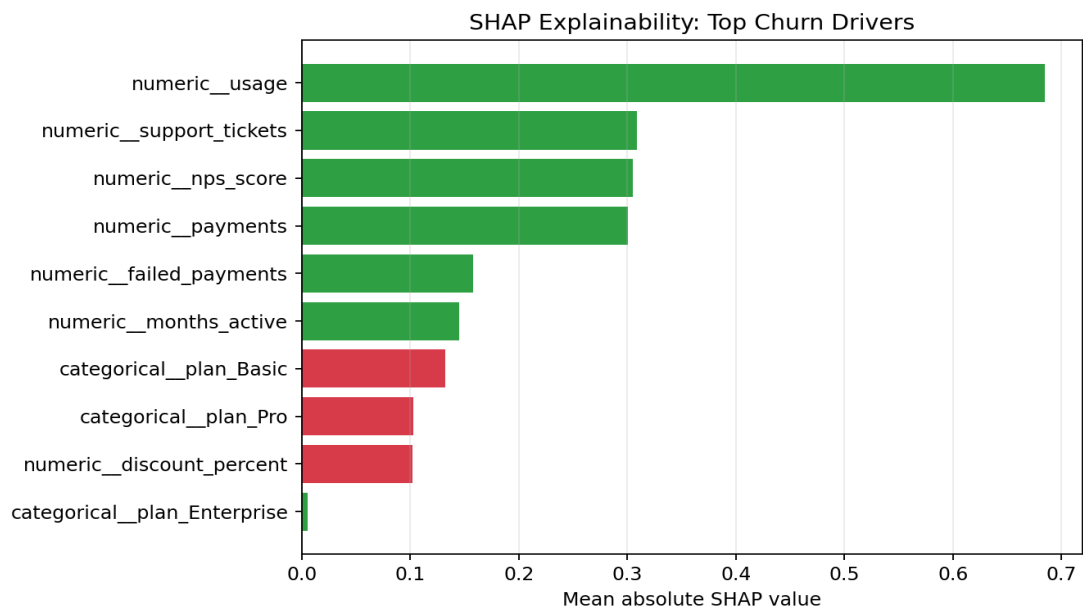
Visual Evidence



Probability Sanity Check



SHAP Explainability



Limitations

This report does not guarantee a specific accuracy number. Results depend on data volume, feature quality, label quality, and whether post-churn leakage columns were excluded before modeling.

Next Steps

Recommended next steps: review leakage warnings, confirm the business definition of churn, enrich weak features, run the action queue, and compare retention outcomes after the next billing cycle.

Scope and Privacy

This package is for a one-time churn analysis and retention report from a provided CSV. API deployment, recurring automation, CRM integration, and dashboard hosting are custom add-ons. Client data should not be committed to a public repository.